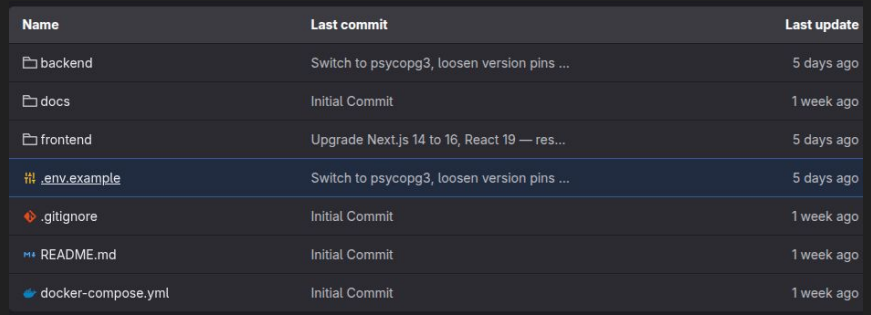
Week 4 Progress

Zackary Siddiqi

Current Progress

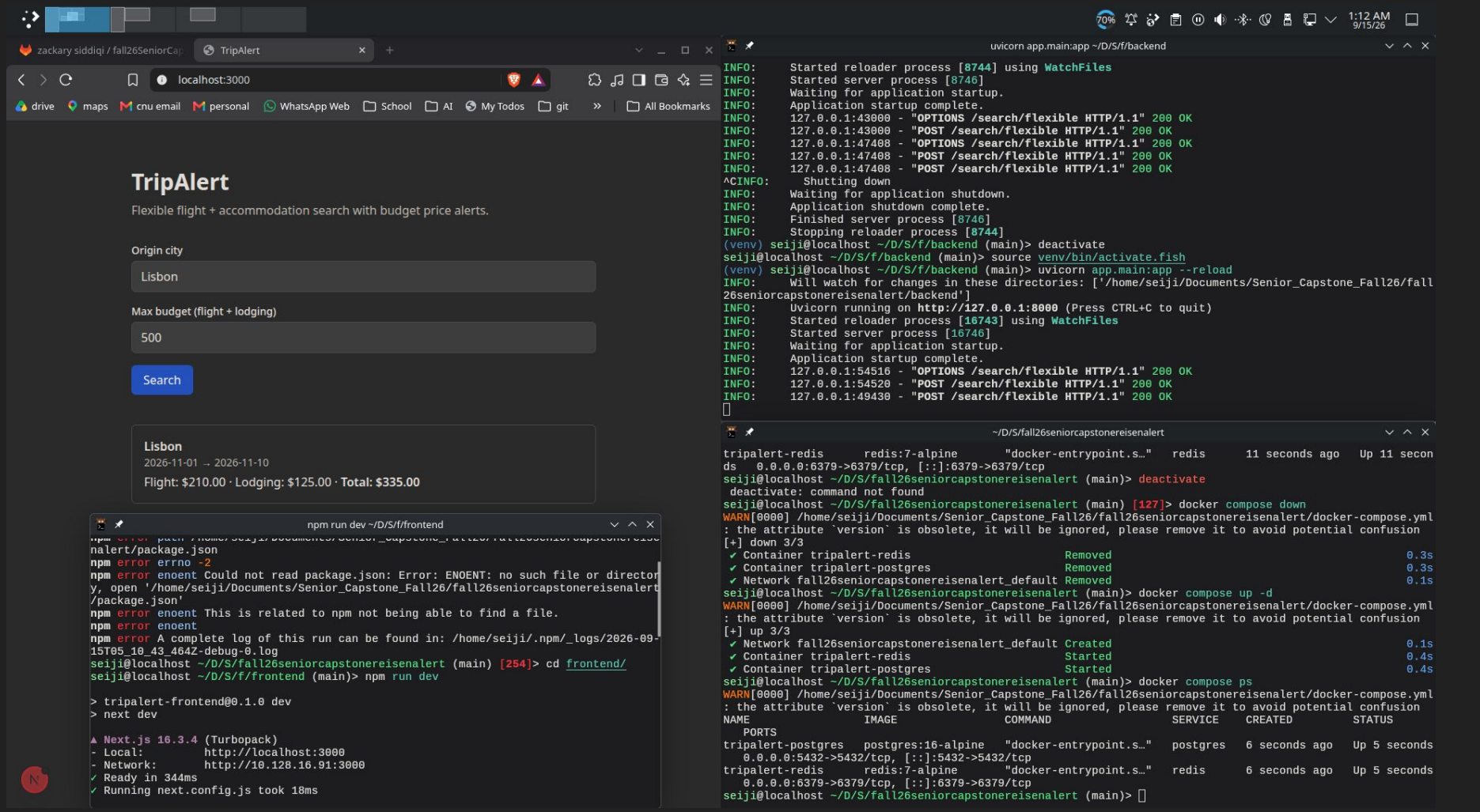
- Created a local Repo and pushed project up to gitlab
- Installed project tools and dependencies
  - Frontend: React, Next.js, Tailwind CSS
  - Backend: FastAPI, Docker, Node.js + npm



Name	Last commit	Last update
backend	Switch to psycopg3, loosen version pins ...	5 days ago
docs	Initial Commit	1 week ago
frontend	Upgrade Next.js 14 to 16, React 19 — res...	5 days ago
.env.example	Switch to psycopg3, loosen version pins ...	5 days ago
.gitignore	Initial Commit	1 week ago
README.md	Initial Commit	1 week ago
docker-compose.yml	Initial Commit	1 week ago

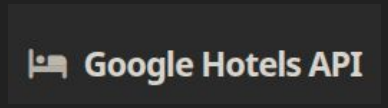
- Performed a test with the app up and running
  - Next.js frontend running with localhost:3000
  - The tests returns hard coded mock data to ensure everything is working

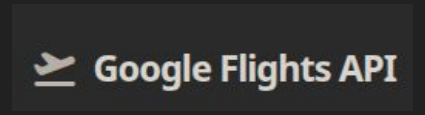
- Key CS Concept: Client-Server Architecture & REST APIs


zackary siddiqi / fall26SeniorCap
TripAlert
localhost:3000
drive
maps
cnu email
personal
WhatsApp Web
School
AI
My Todos
git
All Bookmarks
TripAlert
Flexible flight + accommodation search with budget price alerts.
Origin city
Lisbon
Max budget (flight + lodging)
500
Search
Lisbon
2026-11-01 → 2026-11-10
Flight: $210.00 · Lodging: $125.00 · Total: $335.00
uvicorn app.main:app ~/D/S/f/backend
INFO: Started reloader process [8744] using WatchFiles
INFO: Started server process [8746]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: 127.0.0.1:43000 - "OPTIONS /search/flexible HTTP/1.1" 200 OK
INFO: 127.0.0.1:43000 - "POST /search/flexible HTTP/1.1" 200 OK
INFO: 127.0.0.1:47408 - "OPTIONS /search/flexible HTTP/1.1" 200 OK
INFO: 127.0.0.1:47408 - "POST /search/flexible HTTP/1.1" 200 OK
INFO: 127.0.0.1:47408 - "POST /search/flexible HTTP/1.1" 200 OK
^CINFO: Shutting down
INFO: Waiting for application shutdown.
INFO: Application shutdown complete.
INFO: Finished server process [8746]
INFO: Stopping reloader process [8744]
(venv) seiji@localhost ~/D/S/f/backend (main)> deactivate
seiji@localhost ~/D/S/f/backend (main)> source venv/bin/activate.fish
(venv) seiji@localhost ~/D/S/f/backend (main)> uvicorn app.main:app --reload
INFO: Will watch for changes in these directories: ['/home/seiji/Documents/Senior_Capstone_Fall26/fall26seniorcapstonereisenalert/backend']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [16743] using WatchFiles
INFO: Started server process [16746]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: 127.0.0.1:54516 - "OPTIONS /search/flexible HTTP/1.1" 200 OK
INFO: 127.0.0.1:54520 - "POST /search/flexible HTTP/1.1" 200 OK
INFO: 127.0.0.1:49430 - "POST /search/flexible HTTP/1.1" 200 OK
~/D/S/fall26seniorcapstonereisenalert
tripalert-redis redis:7-alpine "docker-entrypoint.s…" redis 11 seconds ago Up 11 seconds 0.0.0.0:6379->6379/tcp, [::]:6379->6379/tcp
seiji@localhost ~/D/S/fall26seniorcapstonereisenalert (main)> deactivate
deactivate: command not found
seiji@localhost ~/D/S/fall26seniorcapstonereisenalert (main) [127]> docker compose down
WARN[0000] /home/seiji/Documents/Senior_Capstone_Fall26/fall26seniorcapstonereisenalert/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] down 3/3
✔ Container tripalert-redis Removed 0.3s
✔ Container tripalert-postgres Removed 0.3s
✔ Network fall26seniorcapstonereisenalert_default Removed 0.1s
seiji@localhost ~/D/S/fall26seniorcapstonereisenalert (main)> docker compose up -d
WARN[0000] /home/seiji/Documents/Senior_Capstone_Fall26/fall26seniorcapstonereisenalert/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] up 3/3
✔ Network fall26seniorcapstonereisenalert_default Created 0.1s
✔ Container tripalert-redis Started 0.4s
✔ Container tripalert-postgres Started 0.4s
seiji@localhost ~/D/S/fall26seniorcapstonereisenalert (main)> docker compose ps
WARN[0000] /home/seiji/Documents/Senior_Capstone_Fall26/fall26seniorcapstonereisenalert/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
NAME IMAGE COMMAND SERVICE CREATED STATUS PORTS
tripalert-postgres postgres:16-alpine "docker-entrypoint.s…" postgres 6 seconds ago Up 5 seconds 0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp
tripalert-redis redis:7-alpine "docker-entrypoint.s…" redis 6 seconds ago Up 5 seconds 0.0.0.0:6379->6379/tcp, [::]:6379->6379/tcp
seiji@localhost ~/D/S/fall26seniorcapstonereisenalert (main)>
npm run dev ~/D/S/f/frontend
nalert/package.json
npm error errno -2
npm error enoent Could not read package.json: Error: ENOENT: no such file or directory, open '/home/seiji/Documents/Senior_Capstone_Fall26/fall26seniorcapstonereisenalert/package.json'
npm error enoent This is related to npm not being able to find a file.
npm error enoent
npm error A complete log of this run can be found in: /home/seiji/.npm/_logs/2026-09-15T05_10_43_464Z-debug-0.log
seiji@localhost ~/D/S/fall26seniorcapstonereisenalert (main) [254]> cd frontend/
seiji@localhost ~/D/S/f/frontend (main)> npm run dev
> tripalert-frontend@0.1.0 dev
> next dev
▲ Next.js 16.3.4 (Turbopack)
- Local: http://localhost:3000
- Network: http://10.128.16.91:3000
✓ Ready in 344ms
✓ Running next.config.js took 18ms

One Major Issue







- I was originally planning on using Amadeus for Developers free API. However, since last July, it has been fully decommissioned and unavailable for free use.
- **Why was this problematic?** External Flight APIs are the core external dependency of this project
- **Solution:**
  - Found Serp API which covers both Flights and hotels using google flights' search engine data
  - Returns search results into structured JSON Data
  - *Some constraints:* Only one key and limited to 250 free calls/month

Returns search results into Structured JSON Data

